HOW POPULATIONS EVOLVE

Module map

- Module 13: How Populations Evolve
- Connected lab: lab-13_how-populations-evolve.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Allele frequencies as the unit of change

Sources of variation

Hardy-Weinberg as a baseline

Module 13

Five mechanisms of microevolution

Genetic drift and selection

lab-13_how-populations-evolve.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

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Learning objectives

- Define microevolution using allele frequencies.
- Compare the five mechanisms that change allele frequencies.
- Distinguish bottleneck and founder effects.
- Interpret directional, stabilizing, and disruptive selection.
- Use Hardy-Weinberg as a non-evolving baseline.

OBJECTIVE 1
Define microevolution using allele frequencies.

OBJECTIVE 2
Compare the five mechanisms that change allele frequencies.

OBJECTIVE 3
Distinguish bottleneck and founder effects.

OBJECTIVE 4
Interpret directional, stabilizing, and disruptive selection.

OBJECTIVE 5
Use Hardy-Weinberg as a non-evolving baseline.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

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Topic sequence

- Allele frequencies as the unit of change: Microevolution is change in allele frequencies across generations.
- Sources of variation: Mutation creates new variation, while recombination reshuffles existing variation.
- Five mechanisms of microevolution: Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.
- Genetic drift and selection: Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.
- Hardy-Weinberg as a baseline: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

TOPIC 1
Allele frequencies as the unit of change

TOPIC 2
Sources of variation

TOPIC 3
Five mechanisms of microevolution

TOPIC 4
Genetic drift and selection

TOPIC 5
Hardy-Weinberg as a baseline

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

HOW POPULATIONS EVOLVE

Module 13: How Populations Evolve Evidence Map

- Module focus: How Populations Evolve
- Central claim: How Populations Evolve explains allele-frequency change through drift, gene flow, selection, and baseline models.
- Key nodes to connect: Allele frequencies as the unit of change, Sources of variation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-13_how-populations-evolve.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

HOW POPULATIONS EVOLVE

Module 13: How Populations Evolve Reasoning Sequence

- Module focus: How Populations Evolve
- Trace the model: Allele frequencies as the unit of change -> Sources of variation -> Five mechanisms of microevolution -> Genetic drift and selection
- Inputs become outputs: Allele frequencies as the unit of change, Sources of variation -> Define microevolution using allele frequencies., Compare the five mechanisms that change allele frequencies.
- Feedback check: lab-13_how-populations-evolve.md evidence checks whether students can explain

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

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Terms as evidence handles

- Allele frequency: Proportion of a particular allele in a population.
- Microevolution: Allele-frequency change across generations.
- Mutation: DNA sequence change that can create new alleles.
- Gene flow: Movement of alleles between populations.
- Genetic drift: Random allele-frequency change, strongest in small populations.
- Bottleneck effect: Drift after a sharp population reduction.

ALLELE FREQUENCY
Proportion of a particular allele in a population.

MICROEVOLUTION
Allele-frequency change across generations.

MUTATION
DNA sequence change that can create new alleles.

GENE FLOW
Movement of alleles between populations.

GENETIC DRIFT
Random allele-frequency change, strongest in small populations.

BOTTLENECK EFFECT
Drift after a sharp population reduction.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-13_how-populations-evolve.md
- Lab evidence should test or illustrate: Five mechanisms of microevolution
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Allele frequencies as the unit of change

EVIDENCE
lab-13_how-populations-evolve.md

CLAIM
Define microevolution using allele frequencies.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: Microevolution is change in allele frequencies across generations.
- Stronger explanation: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Microevolution is change in allele frequencies across generations.

DEEPER
Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

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Module 13: How Populations Evolve Retrieval and Lab Check

- Module focus: How Populations Evolve
- Retrieval prompt: What is an allele frequency?
- Answer check: Define microevolution using allele frequencies.
- Required terms: Allele frequency, Microevolution, Mutation
- Lab connection: lab-13_how-populations-evolve.md; lab-13_how-populations-evolve.md supplies evidence for population evolution evidence from frequency changes.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Which statement best defines Allele frequency in Module 13?
- 2. Which learning goal best supports the topic "Sources of variation"?
- 3. How should lab-13_how-populations-evolve.md support Module 13?
- 4. Which retrieval move best prepares a student for How Populations Evolve?

QUIZ 1
A

QUIZ 2
B

QUIZ 3
C

QUIZ 4
D

Teaching note: Use the quiz as formative feedback. Answer key: 1=A, 2=B, 3=C, 4=D

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Synthesis and exit ticket

- Exit claim: explain allele frequencies as the unit of change using evidence from lab-13_how-populations-evolve.md.
- Must include: Allele frequency, Microevolution, Mutation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Allele frequencies as the unit of change

EVIDENCE
lab-13_how-populations-evolve.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.